DO NOW — RECALL

Entity relationship diagrams

Name: _____ Class: _____ Date: _____

1. Where is a FK stored in a one-to-many relationship?

2. What does PK stand for?

3. What is an associative entity?

4. A BST root20 has left10 and right30; insert 25. Describe comparisons and placement.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

On the many side.

2. Model answer

Primary key.

3. Model answer

Bridge/junction entity resolving many-to-many.

4. Model answer

25>20 right to30; 25<30 left child of30.

Vocabulary — know these terms

ERD — Entity relationship diagram.

cardinality — Number of related instances.

junction table — Table linking M:N records.

one-to-many — One entity instance can have several related rows.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model answer

each enrolment references exactly one pupil and course.

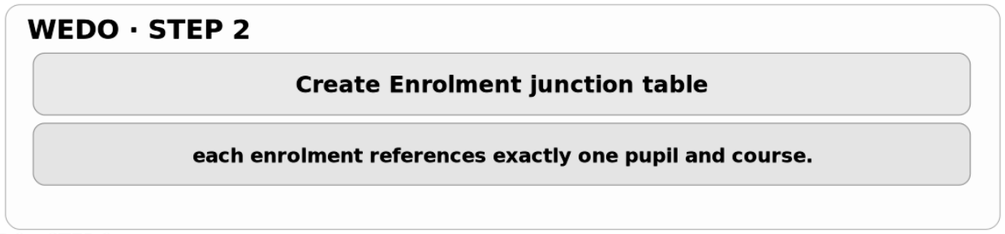
IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Model a pupil taking several courses and each course having several pupils.

Create Enrolment junction table



WEDO · STEP 2

NEW LEARNING

Read, annotate and explain

Understand

ERDs model entities and relationship cardinalities before implementing tables.

Apply

One-to-many relationships are typically represented with a foreign key on the many side.

Evaluate

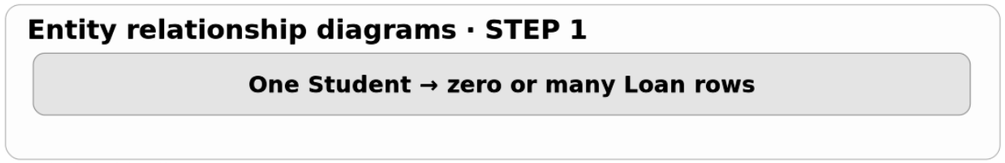
Many-to-many relationships require an associative/junction table when translated into a relational schema, often with two foreign keys.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

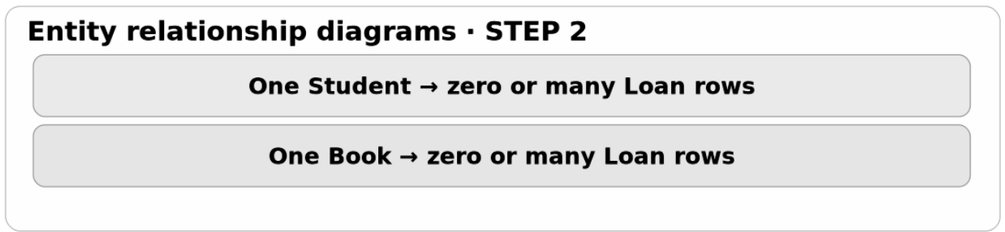
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



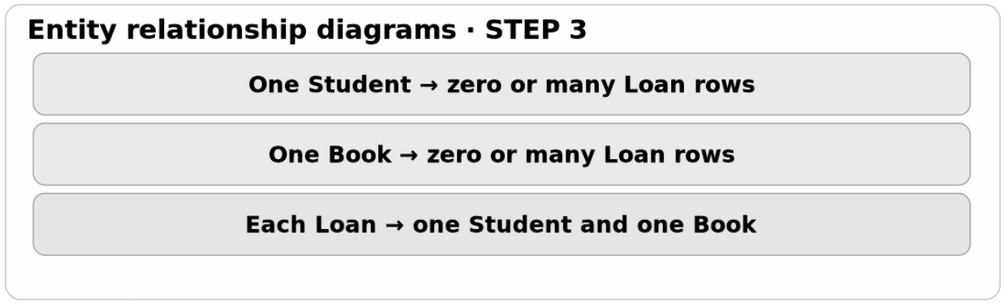
Entity relationship diagrams · STEP 1

New Learning



Entity relationship diagrams · STEP 2

New Learning



Entity relationship diagrams · STEP 3

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A many-to-many relationship is usually represented in relational tables by placing one simple FK on either table.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

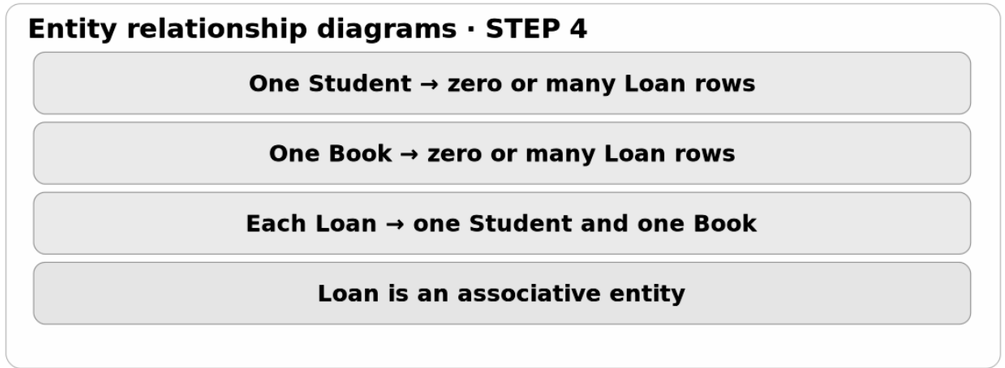
Question 3

Resolve Student many-to-many Course into relational tables; describe keys and integrity. [6]

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Entity relationship diagrams · STEP 4

I DO — WORKED EXAMPLE

Follow the worked method

Model Author and Book when one author writes multiple books, each book has one primary author.

Predict first

Each Book row stores one AuthorID

Model answer

several books can refer to the same author.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State meanings of 1:1,1:M and M:N cardinalities. [3]

Question 2

Draw/describe an ERD for Customer and Order with one customer placing many orders. Give foreign key. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first Create Enrolment junction table

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Model Author and Book when one author writes multiple books, each book has one primary author.

Each Book row stores one AuthorID



IDO · STEP 1

Model Author and Book when one author writes multiple books, each book has one primary author.

Each Book row stores one AuthorID

IDO · STEP 2

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Model a pupil taking several courses and each course having several pupils.

Create Enrolment junction table



WEDO · STEP 1